In-Context Learning

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Model Setting

Linear Attention Model

$$\mathrm{A} = \frac{1}{\sqrt{D+1}} \mathrm{X}\mathrm{X}^\top \mathrm{W}\mathrm{X} \tag{1}$$

the input matrix X is

$$\mathrm{X} = \begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,N} & x_{1,N+1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{D,1} & x_{D,2} & \cdots & x_{D,N} & x_{D,N+1} \\ y_1 & y_2 & \cdots & y_N & 0 \end{bmatrix} = \begin{bmatrix} \mathrm{X}_0 & \boldsymbol{x} \\ \boldsymbol{y}_0^\top & 0 \end{bmatrix} \tag{2}$$

where $\boldsymbol{y}_0 = \mathrm{X}_0^\top \boldsymbol{w}$ and y_{N+1} is set to zero when X is fed into the model.

Let $\mathrm{C} \equiv \mathrm{X}\mathrm{X}^\top$

$$\mathrm{C} = \begin{bmatrix} \mathrm{X}_0 & \boldsymbol{x} \\ \boldsymbol{y}_0^\top & 0 \end{bmatrix} \begin{bmatrix} \mathrm{X}_0^\top & \boldsymbol{y}_0 \\ \boldsymbol{x}^\top & 0 \end{bmatrix} = \begin{bmatrix} \mathrm{X}_0\mathrm{X}_0^\top + \boldsymbol{x}\boldsymbol{x}^\top & \mathrm{X}_0\boldsymbol{y}_0 \\ \boldsymbol{y}_0^\top \mathrm{X}_0^\top & \boldsymbol{y}_0^\top \boldsymbol{y}_0 \end{bmatrix} \tag{3}$$

the prediction

$$\hat{y} = \mathrm{A}_{D+1,N+1} = \frac{1}{\sqrt{D+1}} \sum_{m,n} \mathrm{C}_{D+1,m} \mathrm{W}_{m,n} \mathrm{X}_{n,N+1} \tag{4}$$

From Linear Attention to Glass Model

$$\hat{y} = \mathrm{A}_{D+1,N+1} = \frac{1}{\sqrt{D+1}} \sum_{m,n} \mathrm{C}_{D+1,m} \mathrm{W}_{m,n} \mathrm{X}_{n,N+1}$$

Let

$$\sigma_i = \mathcal{T}\mathrm{W}_{m,n} \qquad s_i = \mathcal{T}s_{m,n} = \mathcal{T}\mathrm{C}_{D+1,m}\mathrm{X}_{n,N+1}/\sqrt{D+1} \tag{5}$$

where $\mathcal{T}$ is an index mapping $\mathcal{T} : (m,n) \to i$ satisfying $i = (D+1)(m-1)+n$.

So we can rewrite the prediction as $\hat{y} = \sum_i s_i\sigma_i$ and the loss of each sample as

$$\ell = \frac{1}{2}\left(y - \sum_i s_i\sigma_i\right)^2 = \frac{1}{2}\sum_{i,j} s_i s_j \sigma_i\sigma_j - \frac{1}{2}y\sum_i s_i\sigma_i + \frac{1}{2}y^2 \tag{6}$$

let $J_{ij} = -\frac{1}{2}s_i s_j$, $h_i = \frac{1}{2}ys_i$, and omitting the constant term $\frac{1}{2}y^2$, so we get a glass model

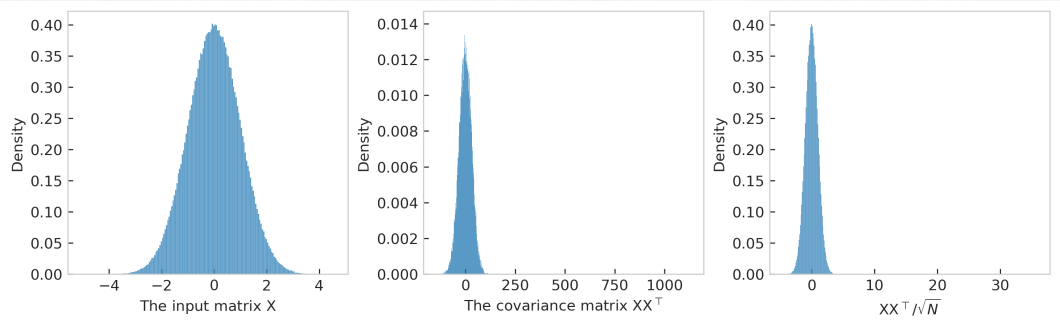
$$\ell = \mathcal{H} = -\sum_{i,j} J_{ij}\sigma_i\sigma_j - \sum_i h_i\sigma_i \tag{7}$$

协方差矩阵的数量级问题

模型

$$\mathrm{A} = \frac{1}{\sqrt{D+1}} \mathrm{X}\mathrm{X}^\top \mathrm{W}\mathrm{X} \tag{8}$$

其中 $1/\sqrt{D+1}$ 仅仅保证 $\mathrm{W}\mathrm{X}$ 是 $\mathcal{O}(1)$ 的。协方差矩阵



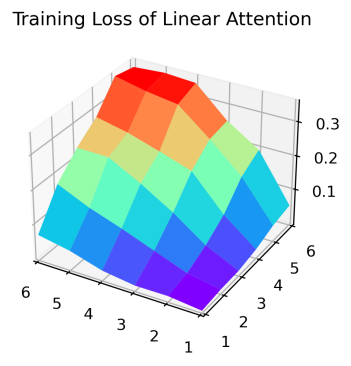
归一化系数如何确定?

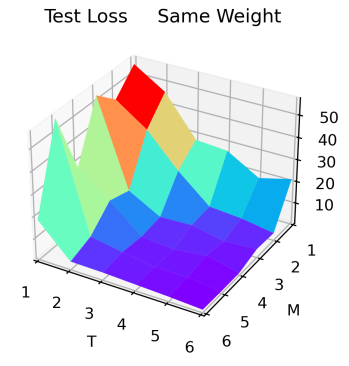
$$\mathrm{A} = \frac{1}{\sqrt{(D+1)(N+1)}} \mathrm{X}\mathrm{X}^\top \mathrm{W}\mathrm{X} \tag{9}$$

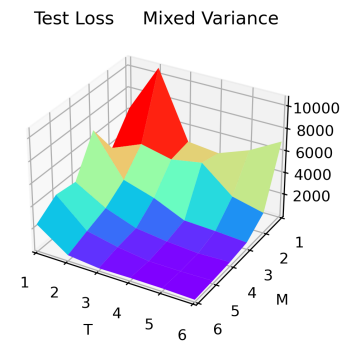
训练结果

四种不同的测试集

- 同权重 $\boldsymbol{w}_{\text{te}} = \boldsymbol{w}_{\text{tr}}$
- 同方差 $\boldsymbol{w}_{\text{te}} \sim \mathcal{N}(0, \sigma_{\text{tr}})$
- 不同方差 $\boldsymbol{w}_{\text{te}} \sim \mathcal{N}(0, \sigma_{\text{te}})$，$\sigma_{\text{te}} \neq \sigma_{\text{tr}}$
- 混合方差 $\boldsymbol{w}_{\text{te}} = [\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \cdots]$，$\mathbf{w}_1 \sim \mathcal{N}(0,1)$，$\mathbf{w}_2 \sim \mathcal{N}(0,2)$，$\mathbf{w}_3 \sim \mathcal{N}(0,3), \cdots$







Linear Attention 测试误差的数值太大了

More Results

Linear Attention 的训练

协方差矩阵

$$\mathrm{C} = \begin{bmatrix} \mathrm{X}_0 & \boldsymbol{x} \\ \boldsymbol{y}_0^\top & 0 \end{bmatrix} \begin{bmatrix} \mathrm{X}_0^\top & \boldsymbol{y}_0 \\ \boldsymbol{x}^\top & 0 \end{bmatrix} = \begin{bmatrix} \mathrm{X}_0\mathrm{X}_0^\top + \boldsymbol{x}\boldsymbol{x}^\top & \mathrm{X}_0\boldsymbol{y}_0 \\ \boldsymbol{y}_0^\top\mathrm{X}_0^\top & \boldsymbol{y}_0^\top\boldsymbol{y}_0 \end{bmatrix} \tag{10}$$

感知机项

$$\mathrm{WX} = \begin{bmatrix} \mathrm{W}_{11} & \mathrm{W}_{12} \\ \mathrm{W}_{21} & \mathrm{W}_{22} \end{bmatrix} \begin{bmatrix} \mathrm{X}_0 & \boldsymbol{x} \\ \boldsymbol{y}_0^\top & 0 \end{bmatrix} = \begin{bmatrix} \mathrm{W}_{11}\mathrm{X}_0 + \mathrm{W}_{12}\boldsymbol{x} & \mathrm{W}_{11}\boldsymbol{x} \\ \mathrm{W}_{21}\mathrm{X}_0 + \mathrm{W}_{22}\boldsymbol{y}_0^\top & \mathrm{W}_{21}\boldsymbol{x} \end{bmatrix} \tag{11}$$

模型的预测值

$$\begin{aligned} \hat{y} = [\mathrm{CWX}]_{D+1,N+1} &= \boldsymbol{y}_0^\top\mathrm{X}_0^\top\mathrm{W}_{11}\boldsymbol{x} + \boldsymbol{y}_0^\top\boldsymbol{y}_0\mathrm{W}_{21}\boldsymbol{x} \\ &= \boldsymbol{w}^\top\mathrm{X}_0\mathrm{X}_0^\top\mathrm{W}_{11}\boldsymbol{x} + \boldsymbol{w}^\top\mathrm{X}_0\mathrm{X}_0^\top\boldsymbol{w}\mathrm{W}_{21}\boldsymbol{x} \\ &= \boldsymbol{w}^\top\left(\mathrm{W}_{11} + \boldsymbol{w}\mathrm{W}_{21}\right)\boldsymbol{x} \end{aligned} \tag{12}$$

其中利用了 $\boldsymbol{y}_0 = \mathrm{X}_0^\top\boldsymbol{w}$ 以及 $\mathrm{X}_0\mathrm{X}_0^\top = \mathbb{I}$。对比训练集的标签 $y = \boldsymbol{w}^\top\boldsymbol{x}$，可得

$$\mathrm{W}_{11} + \boldsymbol{w}\mathrm{W}_{21} = \mathbb{I} \tag{13}$$

一个训练好的模型，应该不与特定的 $\boldsymbol{w}$ 有关，因此我们期望 $\mathrm{W}_{21} = 0$，$\mathrm{W}_{11} = \mathbb{I}$

实验验证

-0.31	-0.75	-0.95	-1.84	-1.04	0.33	0.66
0.14	-0.68	0.73	1.28	1.59	-0.15	-0.52
0.81	1.98	-0.49	0.94	0.92	-1.76	-1.14
0.88	1.24	-0.43	0.91	-1.05	1.46	-0.47
-0.15	0.28	0.86	0.56	0.27	-0.60	-0.69
-0.60	-0.58	1.11	1.60	0.11	-0.36	1.00
0.32	1.26	-0.08	0.59	0.53	-0.58	0.35

The initial weight matrix

0.18	0.00	0.00	-0.00	-0.00	0.00	0.66
0.00	0.17	-0.00	0.00	-0.00	-0.00	-0.52
0.00	-0.00	0.17	-0.00	0.00	-0.00	-1.14
-0.00	0.00	0.00	0.17	0.00	0.00	-0.47
-0.00	-0.00	0.00	0.00	0.18	0.00	-0.69
0.00	-0.00	-0.00	0.00	0.00	0.17	1.00
0.00	-0.00	0.00	0.00	-0.00	-0.00	0.35

The trained weight matrix

Equivalent Model

Linear Attention

$$\hat{y} = \boldsymbol{y}_0^\top \mathrm{X}_0^\top \mathrm{W}_1 \boldsymbol{x} + \boldsymbol{y}_0^\top \boldsymbol{y}_0 \mathrm{W}_2 \boldsymbol{x} \tag{14}$$

Percrptron

$$\hat{y} = \boldsymbol{w}\boldsymbol{x} \tag{15}$$

需要进行进一步的实验

Cavity Method

The probability of each configuration is given by

$$P(\{\sigma\}) = \frac{1}{Z}\,\mathrm{e}^{-\beta\mathcal{H}} = \frac{1}{Z}\prod_{i\neq j}\mathrm{e}^{\beta J_{ij}\sigma_i\sigma_j}\prod_i \mathrm{e}^{\beta h_i\sigma_i} \tag{16}$$

The cavity equation reads

$$\eta_{i\to j}(\sigma_i) = \frac{1}{z_{i\to j}}\mathrm{e}^{\beta h_i\sigma_i}\prod_{k\in\partial i\setminus j}\left[\int \mathrm{d}\sigma_k\;\eta_{k\to i}(\sigma_k)\;\mathrm{e}^{\beta J_{ik}\sigma_i\sigma_k}\right] \tag{17}$$

After each cavity probability $\eta_{i\to j}(\sigma_i)$ has converged to its fixed point, the marginal probability distribution for each variable can be computed as

$$\eta_i(\sigma_i) = \frac{1}{z_i}\mathrm{e}^{\beta h_i\sigma_i}\prod_{j\in\partial i}\left[\int \mathrm{d}\sigma_j\;\eta_{j\to i}(\sigma_j)\;\mathrm{e}^{\beta J_{ij}\sigma_i\sigma_j}\right] \tag{18}$$

Relaxed Belief Propagation

If, by the central limit theorem or whatever, we can assume that $\eta_{i\to j}$ obeys Gaussian distribution approximately, i.e.

$$\eta_{i\to j}(\sigma_i) \simeq \exp\left[\frac{m_{i\to j}}{v_{i\to j}}\sigma_i - \frac{1}{2v_{i\to a}}\sigma_i^2\right] \tag{19}$$

with mean $m_{i\to j}$ and variance $v_{i\to j}$

$$\begin{aligned} m_{i\to j} &= \int \mathrm{d}\sigma_i \eta_{i\to j}(\sigma_i)\sigma_i \\ v_{i\to j} &= \int \mathrm{d}\sigma_i \eta_{i\to j}(\sigma_i)\sigma_i^2 - m_{i\to j}^2 \end{aligned} \tag{20}$$

Expanding to second order in J, we can rewrite Eq.(17) as

$$\begin{aligned} \eta_{i\to j}(\sigma_i) &\simeq \exp\left(\beta h_i \sigma_i\right) \prod_{k\in\partial i\setminus j} \left\{ \int \mathrm{d}\sigma_k\, \eta_{k\to i}(\sigma_k) \left[1 + \beta J_{ik}\sigma_i\sigma_k + \frac{1}{2}\beta^2 J_{ik}^2 \sigma_i^2 \sigma_k^2\right]\right\} \\ &= \exp\left(\beta h_i \sigma_i\right) \prod_{k\in\partial i\setminus j} \left[1 + \beta J_{ik}\sigma_i m_{k\to i} + \frac{1}{2}\beta^2 J_{ik}^2 \sigma_i^2 \left(v_{k\to i} - m_{k\to i}^2\right)\right] \\ &\simeq \exp\left[\beta h_i \sigma_i + \sum_{k\in\partial i\setminus j}\left(\beta J_{ik}\sigma_i m_{k\to i} + \frac{1}{2}\beta^2 J_{ik}^2 \sigma_i^2 v_{k\to i}\right)\right] \end{aligned} \tag{21}$$

Comparing Eq.(19) and Eq.(21), one obtains the following self consistency equations for the means and the variances of the Gaussian marginals:

$$\frac{m_{i\to j}}{v_{i\to j}} = \beta h_i + \beta \sum_{k\in\partial i\setminus j} J_{ij} m_{j\to k} \tag{22a}$$

$$\frac{1}{v_{i\to j}} = -\beta^2 \sum_{k\in\partial i\setminus j} J_{ij}^2 v_{j\to k} \tag{22b}$$